DATAMATICS AI ENGINEERING HACKATHON 2026 — TECHNICAL DOCUMENTATION

DATADYNAMOS SYSTEM ARCHITECTURE SPECIFICATION

Comprehensive End-to-End Pipeline, Multi-Engine OCR Orchestration & Rule Engine Blueprint

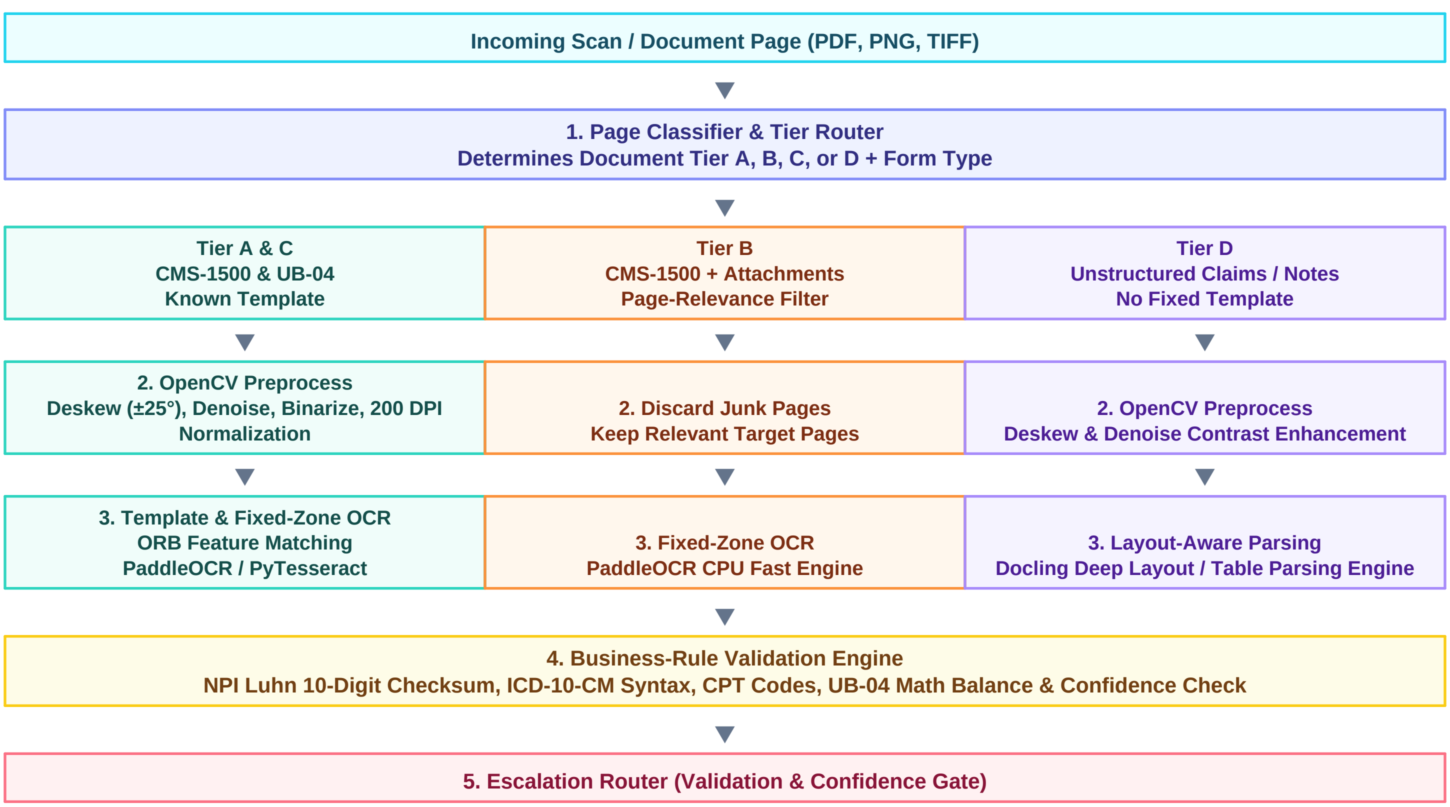
1. System Architectural Overview & Design Philosophy

DataDynamos is an enterprise-grade, zero-retraining platform engineered to process healthcare forms (**CMS-1500**, **UB-04**, **Unstructured Medical Claims**), Commercial Invoices, and Legal Contracts at a target scale of **100 Million pages per year** at **$0.000375 / page average cost** with **93.5% Straight-Through Processing (STP)**.

- **CPU-First Cost Optimization**: 97% of machine-printed forms run on zero-cost, C++ CPU-bound OCR engines (PaddleOCR/PyTesseract at $0.0002/pg), reserving expensive Vision-Language Models (Qwen3-VL-235B at $0.0030/pg) only for low-confidence noisy scans.
- **Deterministic Compliance Primacy**: Hard Python rule engines (NPI Luhn 10-digit checksums, ICD-10-CM format audit, UB-04 revenue math balancing) take precedence over LLM output, preventing AI hallucination in financial/medical claims.
- **Zero-Retraining Prompt Memory HITL Loop**: Inline human corrections captured in the React inspector persist to data/feedback_memory.json and inject learned prompt memory into future LLM extractions without model weight retraining.
- **Spatial Bounding-Box JSON Feeding**: Raw OCR outputs are serialized into rich spatial JSON (bounding box coordinates + Markdown tables) fed into OpenRouter LLMs via the LangExtract framework for high-precision field extraction.

2. End-to-End System Architecture Flowchart

The following multi-tier processing flowchart illustrates document classification, dynamic OpenCV preprocessing, template/anchor vs layout OCR parsing, deterministic rule validation, VLM escalation routing, and the self-learning HITL feedback memory queue:



Pass (Conf >= 80% & Rules Pass) ▼ 6. Structured JSON Aggregation JSON Payload + Audit Trail	Fail / Low Confidence (< 80%) ▼ 5a/b. Vision AI Escalation Qwen3-VL-235B / GPT-4o VLM

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7. Self-Learning HITL Feedback Queue
Human Operator Field Corrections ➔ Persists to data/feedback_memory.json ➔ Injects Prompt Memory

3. Detailed Stage-by-Stage Implementation Matrix

Stage	Pipeline Layer & Module	Input ➔ Core Operation ➔ Output Artifact	Latency & Cost
Stage 1	**Pre-Scan Quality Engine** app/pipeline/prescan.py	Incoming PDF/Scan ➔ OpenCV Deskew (±25°), DPI Normalization (200 DPI), Blur/Contrast Check ➔ Cleaned BGR Raster & Quality Report	45 ms $0.0000
Stage 2	**Format AI Classifier** app/pipeline/classifier.py	Raster Page ➔ Layout & Keyword Heuristics ➔ Tier A (CMS-1500 Single), Tier B (CMS-1500 Multi), Tier C (UB-04), Tier D (Unstructured)	15 ms $0.0000
Stage 3	**Multi-Engine OCR Orchestrator** app/pipeline/ocr/*	Clean Raster ➔ PaddleOCR CPU / PyTesseract / Docling (Tables) / Qwen3-VL Escalation (Noise <80%) ➔ Block OCR JSON + `bbox` Coordinates	320 ms $0.0002
Stage 4	**Spatial LLM Field Structurer** app/pipeline/structuring.py	Block OCR JSON ➔ LangExtract Spatial JSON Serialization ➔ OpenRouter LLMs (DeepSeek-v4 / GPT-4o) ➔ Structured JSON + Grounding Map	680 ms $0.00012
Stage 5	**Deterministic Rule Audit Engine** app/rules/*	Structured JSON ➔ NPI Luhn Checksum, ICD-10 Audit, CPT Code Check, UB-04 Revenue Line Math Balance ➔ Audit Check Results Array	10 ms $0.0000
Stage 6	**Autonomous Decision Agent** app/pipeline/agent.py	Audit Checks + LLM Reasoning ➔ Reconcile Code Rules & LLM Judgment ➔ Verdict: `approve` | `needs_review` | `flag`	350 ms $0.000055
Stage 7	**Self-Learning HITL Feedback Loop** app/routes/documents.py	Human Operator Corrections ➔ Persist to `data/feedback_memory.json` ➔ Inject Dynamic Exemplars in Structuring Prompts	Async $0.0000

4. Multi-Engine OCR Orchestration & Dynamic Escalation Logic

To achieve sub-$0.0004 per page costs across 100M pages, DataDynamos implements a tiered OCR orchestration routing architecture:

- **PaddleOCR (PP-OCRv4 CPU)** (*$0.0002 / page*): Primary default engine for standard machine-printed grid forms (CMS-1500). Executes in C++ on CPU with ~1.2s per page latency and zero commercial API costs.
- **PyTesseract (v5.3)** (*$0.0001 / page*): Ultra-fast fallback CPU engine for standard monospaced text lines and fixed-layout institutional forms (UB-04).
- **Docling Deep Layout Parser** (*$0.0003 / page*): Specialized layout analysis model for multi-column documents, complex tables, and Markdown tabular extraction.
- **Qwen3-VL-235B Vision AI** (*$0.0030 / page*): Multimodal Vision-Language Model invoked over OpenRouter *conditionally* when OCR confidence drops below 80% or severe physical skew/staining is detected.

5. Deterministic Rule Audit Engine & ANSI Healthcare Compliance

Rule Identifier	Target Document Tier	Validation & Compliance Audit Logic	Severity & Action
billing_npi_luhn [ANSI B1]	CMS-1500 / UB-04	Validates 10-digit National Provider Identifier using US prefix `80840` and ANSI Luhn checksum algorithm.	**Hard Failure** (Forces `flag`)
icd10_cm_format [ANSI C1]	CMS-1500 / UB-04	Verifies ICD-10-CM diagnosis code syntax (1 letter + 2 digits + optional decimal + 1-4 characters).	**Hard Failure** (Forces `flag`)
revenue_charges_balance [ANSI D2]	UB-04 Institutional	Verifies that the sum of individual revenue line item charges (Box 47) equals the total billed amount.	**Hard Failure** (Forces `flag`)
patient_identity_match [ANSI A1]	All Healthcare Claims	Ensures Patient Name, DOB, and Policy ID exist and match cross-field references.	**Review Gate** (Caps at `needs_review`)
extraction_confidence	Cross-Cutting (All)	Ensures average extraction confidence across all fields exceeds 80%.	**Review Gate** (Caps at `needs_review`)

6. Technology Stack & Component Mapping

Layer	Technology / Framework	Role & Technical Implementation
Backend API	FastAPI + Uvicorn	Asynchronous REST API with auto-generated OpenAPI docs (`:8000`).
Package Manager	Astral `uv`	Lightning-fast Python dependency locking and virtualenv sync.
Computer Vision	OpenCV 4.x + PyMuPDF	Deskew, binarization, DPI scaling, and raster rendering.
OCR Engines	PaddleOCR + Tesseract + Docling	Multi-engine CPU OCR and layout parsing orchestration.
LLM Structuring	LangExtract Framework	Structured JSON extraction over OpenRouter LLMs (DeepSeek-v4, GPT-4o).
Frontend SPA	React 18 + Vite + TypeScript	Glassmorphism dashboard with split inspector and bounding box highlights.
UI Components	TailwindCSS v4 + Lucide Icons	Modern responsive UI design system with interactive inspector tabs.
Reporting & Export	ReportLab + Pandas + OpenPyXL	Automated PDF specification and Excel benchmark generation.

7. Enterprise Deployment & Scalability Architecture

To support **100 Million pages per year** (equivalent to 70.4 pages per second across a 100-worker cluster):

- **Stateless Worker Nodes**: FastAPI backend workers execute statelessly in Docker containers, allowing horizontal auto-scaling on Kubernetes.
- **Docker Compose Infrastructure**: Unified multi-stage containerization orchestrating backend API (`:8000`) and Vite frontend (`:5173`).
- **Zero-GPU Operational Flexibility**: Standard claim forms run 100% on CPU worker nodes, eliminating expensive GPU cloud infrastructure dependencies.